

FOREST RESOURCES

A forest is a biotic community which is predominantly composed of trees, shrubs or vegetation. Approximately one third of the earth's total land area is covered by forests. Forests are the valuable wealth of a country. They are store houses of biodiversity and provide important environmental services to mankind.

To offer these services, forests have following three types of functions:

1. Productive Functions: Includes the production of timber, bamboos, food and a wide variety of compounds, such as resins, essential oils, latex, pharmaceuticals etc.

The forest products, collected by people include food like fruits, roots, herbs and medicinal plants. People depend on fuel wood to cook food; collect fodder for domestic animals and cut building material for housing; collect medicinal plants that have been known for generations to treat several ailments.

2. Protective Functions: These functions include conservation of soil and water, prevention of drought and protection against wind, cold, radiation, noise and smells.

Forests control the flow of water in streams and rivers. Forest cover reduces the surface run off of rainwater and allows groundwater to be stored. Forests also prevent the erosion of soil.

3. Regulative Functions: Includes absorption, storage and release of gases like CO₂ and O₂, water, mineral elements and radiant energy. Floods, draughts and global biogeochemical cycles, particularly of carbon is also regulated by forests. The regulative functions of forests improve atmospheric and temperature conditions.

Forest Area in India:

India had abundant forests in ancient times. But increase in population, lack of foresight and unplanned felling have drastically reduced the area and quality of forests in India today. At the beginning of the 20th century about 30 per cent of land in India was covered with forests. But by the end of the 20th century the forest cover was reduced to 19.4 per cent.

Deforestation

Destruction of forests is a formidable threat to the quality of life, country's economy and future development.

Causes of Deforestation

The main causes of deforestation are as follows:

1. **Shifting Cultivation (Jhum):** The jhum or shifting cultivation is a traditional agro-forestry system widely practiced in the north eastern region of our country. It involves felling and burning of forests followed by cultivation of crops for few years and abandoning cultivation to allow forest's regrowth.

2. **Explosion of Human Population:** Due to overpopulation of human beings, requirement of timber, fuel, paper, wood etc. has been increased. Man has cleared large areas of forests for agriculture, housing, factories, roads and railway tracks.

3. **Demand of Wood for Industries:** Wood is used for several industrial processes such as making boxes, crates, packing cases, furniture, match boxes, paper, plywood etc.

4. **Construction of Roads:** Construction of roads along the mountains which cover nearly 30,000 km in ecologically fragile area is another cause of forest degradation.
5. **Mining Operations:** Mining operations have a serious impact on forest areas. Large areas of forest are cleared and laid barren as a result of open cast mining of mica, coal, manganese, limestone etc.
6. **Overgrazing:** Overgrazing of forests by livestock has resulted in large scale degradation of forests.
7. **Pests:** Many kinds of insects and pests destroy trees by eating up leaves, boring into shoots and spreading diseases.
8. **Fires:** Fire is the worst enemy of forest. It destroys the full grown trees, seedlings, seeds and even humus. It also causes large scale damage to animal life.
9. **Weather:** Extreme weather conditions such as frost, storms and heat also destroy forests.
10. **Dams and Hydroelectric Projects:** Dams, reservoirs and hydroelectric projects submerge forest, displace local people cause water logging and siltation and may result in earthquakes.

Effects of Deforestation

1. When forest is removed, the amount of runoff water flowing into rivers and streams increases causing frequent floods.
2. Deforestation results in increased soil erosion and decreased soil fertility.
3. Deforestation causes the extinction of forest dwelling plant, animal and microbial species resulting into a loss of genetic resources.
4. Deforestation induces regional and global climate change.
5. The pattern of rainfall has changed in deforested areas. The rainfall has declined and droughts have become common.
6. Deforestation also contributes to global warming by releasing stored carbon into the atmosphere as carbon dioxide, which is a greenhouse gas.

Forest Conservation and Management

1. A tree removed from the forest for any purpose must be replaced by a new tree.
2. Afforestation should be done in areas unfit for agriculture, along highways and river, around playgrounds and parks. A special programme of trees plantation called **Van Mahotsava** is held in the months of August and February, every year in our country.
3. Maximum economy should be observed in the use of timber and fuel wood by minimizing the wastage.
4. The use of fire wood should be discouraged and alternative source of energy for cooking such as biogas, natural gas etc. should be made available.

5. Forest should be **protected from fire**. Modern firefighting equipment should be used to extinguish accidental forest fire.
6. Pests and diseases of forest trees should be controlled by **fumigation and aerial spray of fungicides and through biological method of pest control**.
7. **Grazing of cattle's in the forests should be discouraged.**